David Phan

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Summary of Qualifications

Programming languages/frameworks: Rust, Java, Python, C, C++, Scala, HTML, CSS, Javascript, Typescript, React, Guice, Dagger DI, Bash/Command Line scripting, Linux
Version control/Workflow optimization software: Git, Github, Git Actions, Docker Compose
Database and Cloud service software: AWS, Azure, MongoDB, PostgreSQL
Leadership/Project Management: Engineering Ambassador at UW, ECE Tutor at UW, Officer for Music for Charity at UW, School Music Director, Communications Lead at Washington Aerospace Scholars (WAS)

Education

University of Washington, Seattle Sept. 2020 - June 2024
Bachelor of Science in Electrical and Computer Engineering (ECE) GPA: 3.87, Cum Laude

Experiences and Activities

Software Development Engineer (SDE) II - Amazon Sept. 2024 - Present

- Operated Amazon's core Java/Guice based distributed media ingestion REST API service with regular oncall rotations and peak event preparation.
- Modernized Amazon's content ingestion by adding support for modern media types, like HEIF, and rebuilding components of Amazon's content ingestion platform in Rust for lower latency and memory efficient file type operations, reducing monthly incompatibility ingestion related errors by 32%.
- Streamlined team operations by leading monthly bug bash sessions, onboarding new team members to our team oncall rotations, and migrating our team to new task management tooling.
- Migrated legacy bare-metal deployment infrastructure to CDK based AWS cloud infrastructure.
- Recognized as a top 5 leader within my org in pushing Amazon's leadership principles, leading to an accelerated promotion timeline.

Software Development Engineer (SDE) I Intern - Amazon Jun. 2023 - Sept. 2023

- Spearheaded an internal full-stack application project to aid ML research teams in processing datasets.
- Designed and implemented both a React/Typescript front-end and a REST API using AWS API Gateway, Lambda, S3, DynamoDB, and Amazon internal tools.
- Deployed applications through CI/CD frameworks using Git, internal pipeline tools, and AWS Pipelines.
- Collaborated with a team of developers to generate and meet design goals using the Agile Methodology.

Software Development Intern - Makeability Lab, Project Sidewalk Mar. 2022 – Jun. 2024

- Maintained and updated the Project Sidewalk API and website, which tracks and updates geographical information on pedestrian accessibility issues in current city infrastructure.
- Collaborated with other software developers to resolve operational tickets and implement feature requests for both the Scala, PostgreSQL backend and frontend UI/UX.

Undergraduate Researcher - Theberge (BCME) Lab Jul. 2022 – Jun. 2024

- Automated data-analytics and image processing tasks by coding python-based desktop applications.
- Designed and prototyped microfluidic and biomedical devices for research and medical purposes using CAD software, 3D printing and Milling devices, and microcontroller circuits.
- Built and optimized website features to represent lab research and expand study group participation.